



IOT-ENABLED SMART WASTE COLLECTION SYSTEM WITH COMMUNITY SHARING AND REAL-TIME ROUTE OPTIMIZATION

Requesting Ethical Clearance for Municipal Data Collection

RAKK RANASINGHE (D/BIT/22/0039)

Supervised By: Mr. RPS Kathriarachchi

Research Problem & Justification

Research Problem:

Waste collection inefficiencies in Sri Lanka, such as **fixed schedules, lack of real-time tracking, and low public involvement**, lead to **delays, high costs, overflowing bins, and environmental hazards**.

Key Challenges:

- **Unoptimized Routes:** Causes delays and fuel wastage.
- **No Real-Time Tracking:** Bins and trucks remain unmonitored.
- **Loss of Reusable Waste:** No community-sharing system.
- **Environmental & Health Risks:** Overflowing waste leads to pollution and disease.
- **Low Public Participation:** No platform for citizen involvement.

Justification:

This project integrates **IoT, AI, GIS, and mobile/web applications** to:

- **Optimize waste collection routes**, reducing costs and emissions.
- **Enable real-time tracking** of waste bins and trucks for better responsiveness.
- Encourage **community sharing**, minimizing unnecessary waste.
- Enhance **citizen participation via a mobile and web platform**.
- Promote sustainability by reducing **waste overflow and pollution**.

Data Collection & Testing Process

Data Required:

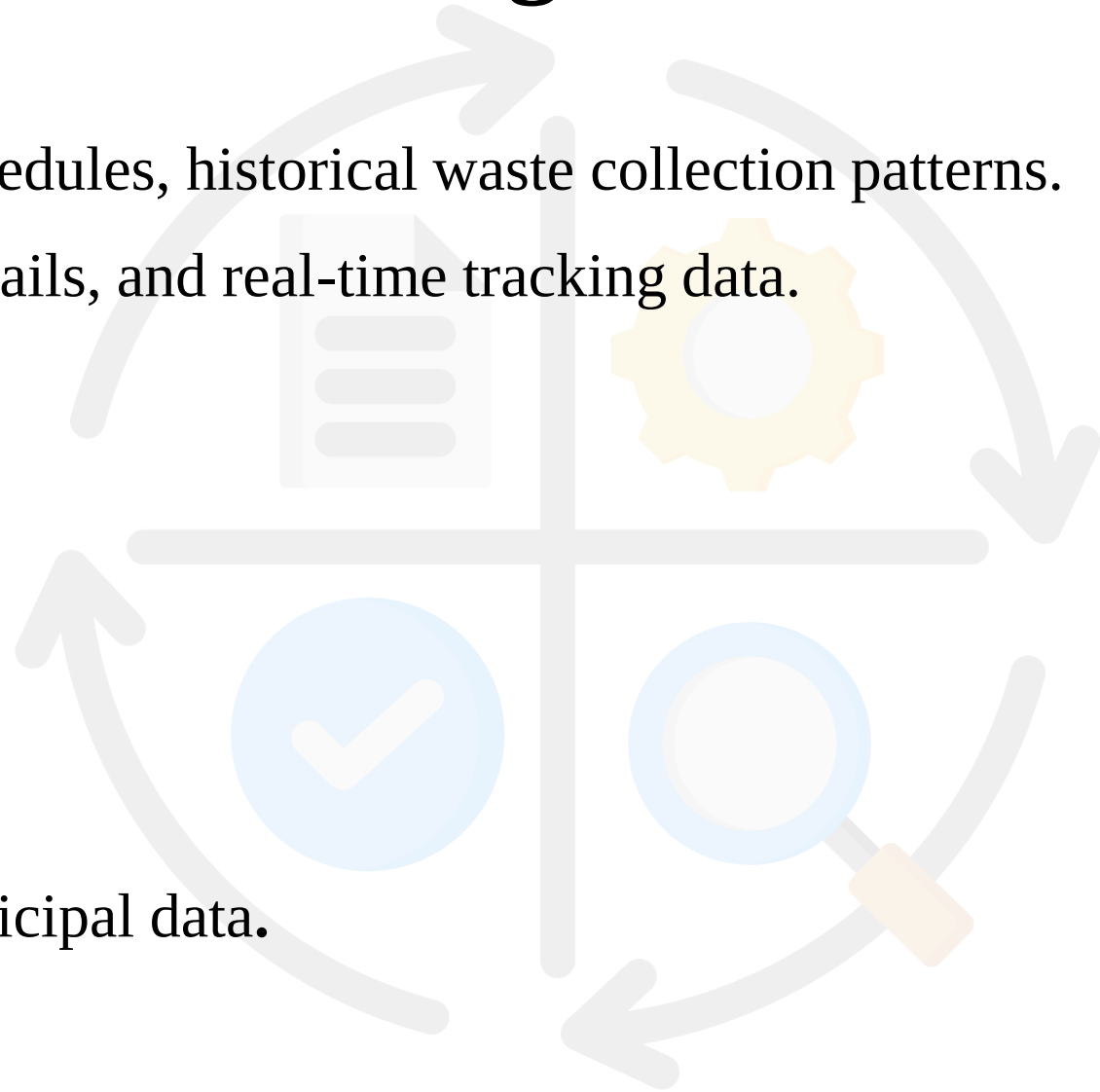
- Route details, waste collection schedules, historical waste collection patterns.
- Municipal reports, bin location details, and real-time tracking data.

Testing Participants:

- Garbage truck drivers
- Municipal administrators
- Residential users.

Testing Approach:

- Simulated environment using municipal data.



Data Confidentiality

Data Protection Measures:

- Municipal reports and historical data will be anonymized.
- Access will be restricted to authorized personnel only.

Security Measures:

- Encryption applied for storage and transmission.
- Controlled access levels for different stakeholders.

Compliance:

- No personal data collection from residents; municipal data is used strictly for research purposes.

Ethical Approval Request & Conclusion

Why is ethical approval needed?

- Ensures responsible use of municipal data and privacy protection.
- Verifies compliance with ethical standards for testing with municipal stakeholders.

Final Request:

- Seeking approval to access municipal reports, route data, and waste collection schedules.
- Testing will be conducted responsibly with garbage truck drivers, municipal admins, and residents.

THANK YOU !